Homework #4

Due: 11:59pm, February 16

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1. **[Sampling]**
   Let $z(t)$ be a continuous-time signal whose CTFT $Z_{\mathsf{c}}(j\Omega)$ satisfies

$$Z_{\mathsf{c}}(j\Omega) = \begin{cases} \frac{|\Omega|}{50\pi} & 0 \le |\Omega| < 50\pi \\ 2 - \frac{|\Omega|}{50\pi} & 50\pi \le |\Omega| < 100\pi \\ 0 & \text{else.} \end{cases}$$

   Let $x(t) = z(t)\cos(300\pi t)$.

   (a) Plot $|X_{\mathsf{c}}(j\Omega)|$. Label both the horizontal axis and the vertical axis.

   (b) Design a bandpass sampling system that enables the exact reconstruction of $x(t)$ from uniform samples. Determine the index of half Nyquist zones and sampling rate $1/T$.

   (c) Let

$$x[n] = x\,(Tn)$$

   for the sampling period T from part (c). Let $X(e^{j\omega})$ be the DTFT of $x[n]$. Plot $|X(e^{j\omega})|$. Label both the horizontal axis and the vertical axis.

2. **[OS 4.22: Sampling]**
   A continuous-time signal $x_{\mathsf{c}}(t)$ with Fourier transform $X_{\mathsf{c}}(j\Omega)$ given by

$$X_{\mathsf{c}}(j\Omega) = \mathrm{tri}\left(\frac{4\Omega - 3\Omega_0}{\Omega_0}\right) + \mathrm{tri}\left(\frac{4\Omega + 3\Omega_0}{\Omega_0}\right),$$

   where

$$\mathrm{tri}(\Omega) = \begin{cases} 1 - |\Omega|, & |\Omega| < 1 \\ 0 & |\Omega| \ge 1. \end{cases}$$

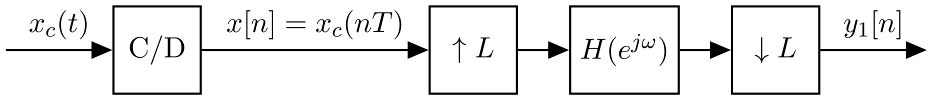
   Then $x_{\mathsf{c}}(t)$ is sampled with sampling period $T = 2\pi/\Omega_0$ to form the sequence $x[n] = x_{\mathsf{c}}(nT)$.

   (a) Sketch $X_{\mathsf{c}}(j\Omega)$ and $X(e^{j\omega})$.

   (b) The signal $x[n]$ is to be transmitted across a digital channel, At the receiver, the original signal $x_{\mathsf{c}}(t)$ must be recovered. Draw a block diagram of the recovery system and specify its characteristics. Assume that ideal filters are available.

   (c) In terms of Ω_0, for what range of values of T can $x_{\mathsf{c}}(t)$ be recovered from $x[n]$?

3. **[OS 4.40]**
   Consider the system shown below.

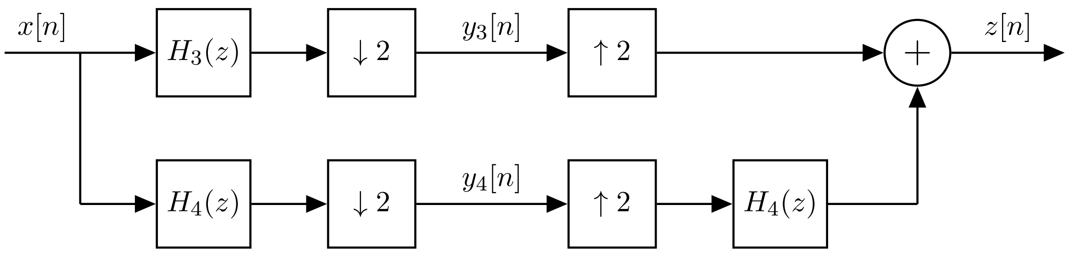


Suppose that $X_c(j\Omega) = 0$ for $|\Omega| \geq \pi/T$ and

$$H(e^{j\omega}) = \begin{cases} e^{-j\omega}, & |\omega| \leq \pi/L, \\ 0, & \pi/L < |\omega| \leq \pi. \end{cases}$$

How is $y[n]$ related to the input signal $x_c(t)$?

4. **[OS 4.46 Part I]**
Consider the following system, where $H_3(z)$ and $H_4(z)$ correspond to LTI systems.

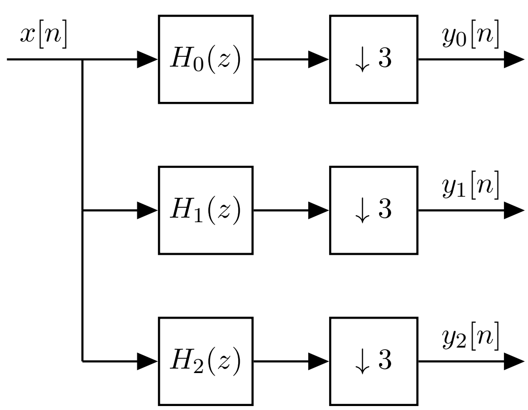


Let $X(e^{j\omega}) = 1 - |\omega/\pi|$ for $-\pi \leq \omega < \pi$, $H_3(e^{j\omega}) = 1$, and

$$H_4(e^{j\omega}) = \begin{cases} 1, & 0 \leq \omega < \pi, \\ -1, & -\pi \leq \omega < 0. \end{cases}$$

Provide the graphs of $X(e^{j\omega})$, $Y_3(e^{j\omega})$, $Y_4(e^{j\omega})$, and $Z(e^{j\omega})$.

5. **[OS 4.46 Part II]**
Consider the system shown below, where $H_0(z)$, $H_1(z)$, and $H_2(z)$ are the system functions of LTI systems.



Assume that $x[n]$ is an arbitrary stable complex signal without any symmetry properties.

(a) Let $H_0(z) = 1$, $H_1(z) = z^{-1}$, and $H_2(z) = z^{-2}$. Can you reconstruct $x[n]$ from $y_0[n]$, $y_1[n]$, and $y_2[n]$? If so, how? If not, justify your answer.

(b) Assume that $H_0(e^{j\omega})$, $H_1(e^{j\omega})$, $H_2(e^{j\omega})$ are as follows:

$$H_0(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \pi/3, \\ 0, & \text{otherwise}, \end{cases}$$

$$H_1(e^{j\omega}) = \begin{cases} 1, & \pi/3 < |\omega| \leq 2\pi/3, \\ 0, & \text{otherwise}, \end{cases}$$

$$H_2(e^{j\omega}) = \begin{cases} 1, & 2\pi/3 < |\omega| \leq \pi, \\ 0, & \text{otherwise}. \end{cases}$$

Can you reconstruct $x[n]$ from $y_0[n]$, $y_1[n]$, and $y_2[n]$? If so, describe a reconstruction scheme. If not, explain why it is not feasible.